Complete Python Syllabus (All Topics)

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1 Python Basics

Introduction to Python

Features of Python

Installing Python (Windows/Mac/Linux)

Python IDEs (IDLE, VSCode, PyCharm)

Writing First Program (`print()`)

Comments (`#`, `''' '''`)

Variables & Constants

Data Types: int, float, string, bool, complex

Type Casting / Type Conversion

Input/Output (`input()`, `print()`)

Keywords & Identifiers

Operators:

Arithmetic, Relational, Logical

Assignment, Bitwise, Membership, Identity

Conditional Statements (`if`, `if-else`, `elif`)

Loops (`for`, `while`, nested loops)

Loop Control Statements: break, continue, pass

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2 Python Functions

Function Definition & Syntax

Parameters & Arguments

Return Statement

Global & Local Variables

Default Arguments

Variable Length Arguments (`*args`, `**kwargs`)

Recursion

Lambda Functions

Map, Filter, Reduce

First Class Functions

Inner / Nested Functions

Decorators

`pass` Statement

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3 Python Data Structures

Strings (Indexing, Slicing, Methods)

Lists (CRUD operations, Methods)

Tuples (Immutable, Methods)

Sets (Operations, Methods)

Dictionaries (Key-Value, CRUD)

Arrays (array module / NumPy arrays)

List Comprehension

Nested Data Structures

Python Collections Module:

Counter

Deque

OrderedDict

DefaultDict

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4 Python Object-Oriented Programming (OOP)

Classes & Objects

Constructors (`__init__`)

Instance & Class Variables

Methods (Instance, Class, Static)

Inheritance (Single, Multiple, Multilevel)

Polymorphism (Method Overloading, Method Overriding)

Abstraction (`abc` module)

Encapsulation (Private / Protected members)

Iterators & Generators

Special / Dunder Methods (`__str__`, `__repr__`, `__len__`, etc.)

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5 Python Exception Handling

Errors vs Exceptions

Try / Except / Else / Finally blocks

Multiple Exception Handling

Raising Exceptions (`raise`)

User-defined Exceptions

Assertions

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6 Python File Handling

Opening Files (`open()`)

Reading Files (`read()`, `readline()`, `readlines()`)

Writing Files (`write()`, `writelines()`)

Working with CSV files (`csv` module)

OS Module (`os`, `os.path`)

Directory Management

`pathlib` Module

Context Managers (`with` statement)

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7 Python Modules & Packages

Importing Modules (`import`, `from ... import`)

Python Standard Library Overview

Creating Custom Modules & Packages

`pip` for Installing Libraries

Virtual Environments (`venv`)

8️⃣ Python Advanced Topics

Decorators

Generators & Iterators

Regular Expressions (`re` module)

Multithreading & Multiprocessing

File Handling Advanced

JSON Handling (`json` module)

Logging (`logging` module)

Date & Time (`datetime` module)

Math & Random modules

Type Hinting / Annotations

9️⃣ Python GUI Programming

Tkinter: Widgets, Layouts, Events, Forms

Kivy: Widgets, Layouts, Multi-touch, KV Language

PyQt / PySide (Optional)

GUI Projects Examples: Calculator, To-Do List, Login Form

🔟 Python Web Development

Introduction to Web Development

Flask Framework Basics (Routes, Templates)

Django Framework Basics (Models, Views, Templates)

APIs with Flask / Django

REST API concepts

CRUD Operations

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11 Python Data Science & AI/ML

NumPy → Arrays, Matrix operations

Pandas → Series, DataFrame, CSV/Excel handling

Matplotlib & Seaborn → Data visualization

Scikit-learn → ML models, Train/Test Split, Evaluation

TensorFlow / PyTorch → Deep Learning basics

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12. Python Projects (Practical)

Beginner Projects:

Calculator (Tkinter/Kivy)

To-Do List App

Contact Book / Address Book

Simple Quiz App

Intermediate Projects:

Student Management System

Expense Tracker

Web Scraper (BeautifulSoup / Scrapy)

Blog Website (Flask/Django)

Advanced Projects:

AI Chatbot

Image Recognition App (TensorFlow)

Data Analysis Dashboard (Pandas + Matplotlib)

E-commerce Website

💡 **Extra Tips:**

Practice small programs daily

Take quizzes & exercises for each topic

Move from basic → intermediate → advanced stepwise

Build projects alongside learning for hands-on experience

Python Practice Exercises

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1. Python Basics

Topics: Variables, Data Types, Operators, I/O, Conditionals, Loops

Practice:

Print your name, age, and city.

Calculate the sum, difference, product, quotient of two numbers.

Check whether a number is even or odd.

Find the largest of three numbers.

Print all numbers from 1 to 50 using a loop.

Print multiplication table of a number.

Reverse a number and check if it's a palindrome.

Count vowels in a string.

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2. Python Functions

Topics: Function syntax, Arguments, Return, Recursion, Lambda, Map/Filter/Reduce

Practice:

Write a function to calculate factorial of a number.

Write a recursive function to find Fibonacci numbers.

Write a lambda function to square a list of numbers using `map()`.

Filter out even numbers from a list using `filter()`.

Write a function to check if a string is a palindrome.

Create a function with `*args` to sum any number of numbers.

Create a function with `**kwargs` to print student info.

3.Python Data Structures

Topics: Strings, Lists, Tuples, Sets, Dictionary, Arrays, List Comprehension

Practice:

Reverse a string using slicing.

Find the largest and smallest element in a list.

Merge two dictionaries.

Count frequency of elements in a list using `Counter`.

Remove duplicates from a list using `set()`.

Create a list of squares using list comprehension.

Find common elements between two lists.

Implement a stack using list.

Sort a dictionary by values.

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4. Python OOP

Topics: Classes, Objects, Inheritance, Polymorphism, Encapsulation

Practice:

Create a `BankAccount` class with deposit and withdraw methods.

Create a `Student` class and store name, roll number, marks.

Implement inheritance: `Person` → `Teacher` & `Student`.

Create a class with private attributes and getter/setter methods.

Implement a custom iterator for a range of numbers.

5.Exception Handling

Topics: Try/Except, Raise, Custom Exceptions

Practice:

Handle division by zero exception.

Handle file not found exception while reading a file.

Create a custom exception for negative input in factorial function.

Handle multiple exceptions in a single try block.

6.File Handling

Topics: Read/Write, CSV, OS, pathlib

Practice:

Read a text file and count words.

Write a list of numbers into a file.

Read a CSV file and print average of a column.

List all files in a directory using `os` module.

Create a folder and write files into it using `pathlib`.

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7. Python Libraries Practice

NumPy / Pandas / Matplotlib / Requests

Practice:

Create a NumPy array and find mean, median, sum.

Create a Pandas DataFrame and perform filtering, sorting.

Plot a line chart and bar chart using Matplotlib.

Fetch data from an API using `requests` and display JSON.

Analyze a CSV file (like sales data) using Pandas.

8 .GUI Practice (Tkinter/Kivy)

Practice:

Create a simple calculator (Tkinter/Kivy).

Create a login form with username & password validation.

Build a To-Do List App with Add/Delete functionality.

Create a student marks management system GUI.

Create a timer/stopwatch app.

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9 .Web Development Practice

Flask / Django

Practice:

Create a basic Flask app showing "Hello World".

Build a CRUD app (Add, View, Update, Delete records) in Flask.

Create a Django project with home page & contact page.

Build a simple blog app with Django (posts, comments).

🔟 Advanced / Projects

Build a command-line Quiz App.

Build a mini Chatbot using Python.

Build a Web Scraper for news headlines using BeautifulSoup.

Build a scientific calculator with GUI.

Analyze stock prices using Pandas & Matplotlib.

Create a simple ML model using Scikit-learn (like Linear Regression).

Build a simple AI Chatbot using TensorFlow/Keras.

💡 **Tips for Practice:**

Solve at least 1–2 exercises daily.

Always try to modify the code to create new variations.

Gradually move from basic → intermediate → advanced → project level.